

Sufficient Statistics for Economic Mobility: When Do Measures Agree?

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UWO & IFS

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A “proliferation of measures”

- Researchers measure economic mobility
 - ... in many outcomes (e.g., incomes, wealth, education, health, ...)
 - ... for many uses (e.g., normative analysis, policy evaluation, validating models, ...)
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- Different measures often produce consistent rankings of mobility (e.g., Katz and Krueger 2017, Berman 2022, Deutscher and Mazumder 2023)
- But not always: long-run US trends (Jácome, Kuziemko & Naidu, 2025); cross-CZ w/in US (Chetty et al., 2014); cross-country (Blanden 2013) ▶ CZs

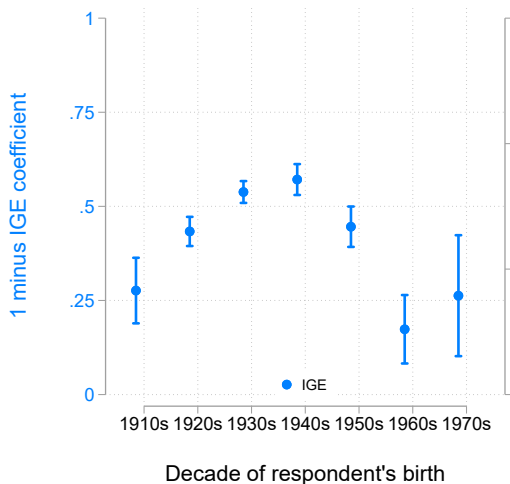
A Puzzle: 20th Century U.S. Intergenerational Mobility

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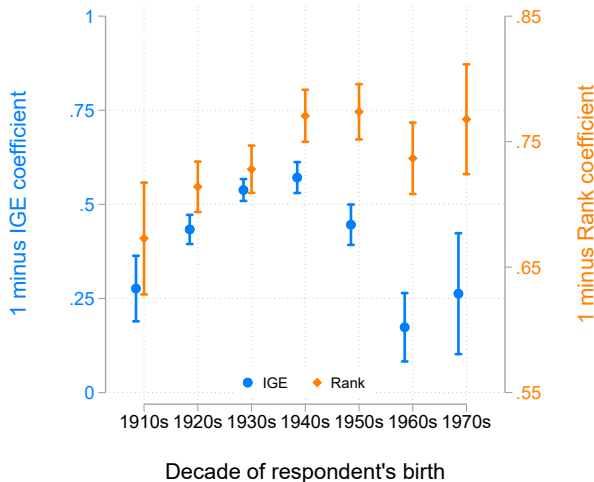
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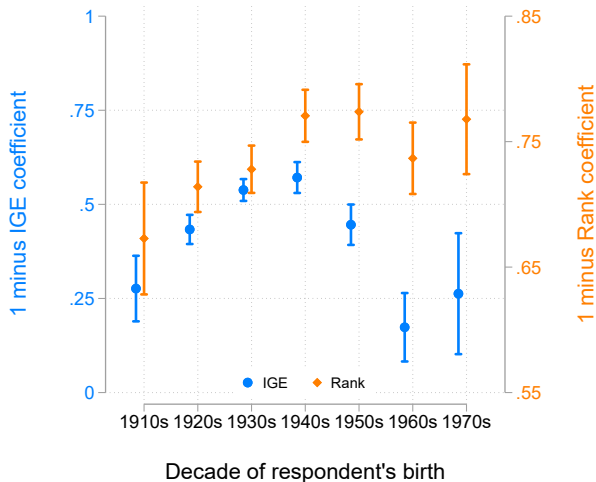
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Two fundamental questions:

1. When *should* we expect measures to agree?
2. What does disagreement *tell us*?



This Paper Provides Answers

1. Agreement of measures depends on single property of parent-child joint distribution
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 - Observe ranking in one measure, know all rankings: measures as ordinal substitutes
 - Fixing marginal distributions of income $\rightarrow$ relative mobility
3. Changes in any mobility measure = rank-dependence + changing inequality
 - Rank-based measures: only rank-dependence
 - Level-based (incl. IGE): both $\rightarrow$ potential for disagreement
 - Disagreement reflects:
 - i. different weightings over regions in joint distribution
 - ii. different sensitivity to changing inequality

Contributions

1. Unifying framework w/ **Concordance** as sufficient statistic for mobility measurement
 - Adds to measure comparisons (Deutscher, Mazumder; Berman), axiomatic approaches (Shorrocks; Fields, Ok; D'Agostino, Dardanoni; Cowell, Flachaire), dual of 2D-inequality problem (Atkinson, Bourguignon; Dardanoni)
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3. Microfounded w/ structural interpretation in Becker-Tomes-Loury models
 - Sorting measures NAM vs PAM in matching (Anderson, Smith; Gola; Boerma, Tsyvinski, Wang, Zhang) and mating (Chiappori, Costa Dias, Meghir, Zhang)

Setup

Separating Dependence from Marginals: Copulas

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Sklar's (1959) Theorem: Any joint distribution decomposes as:

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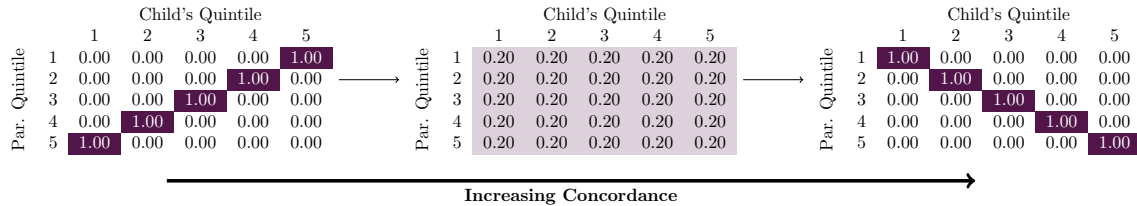
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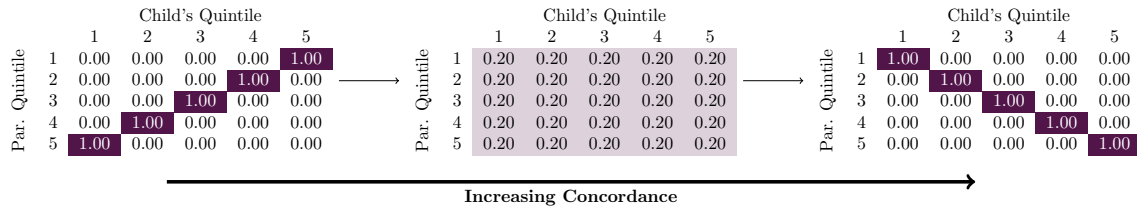
where $C : [0, 1]^2 \rightarrow [0, 1]$ is the **copula** or joint distribution of ranks

- Copula encodes *all* rank-dependence between generations $\perp$ of marginals ► Mass points
- Relative mobility measures alternative summaries of copula

Concordance Order: Visual Intuition ▶ Example II

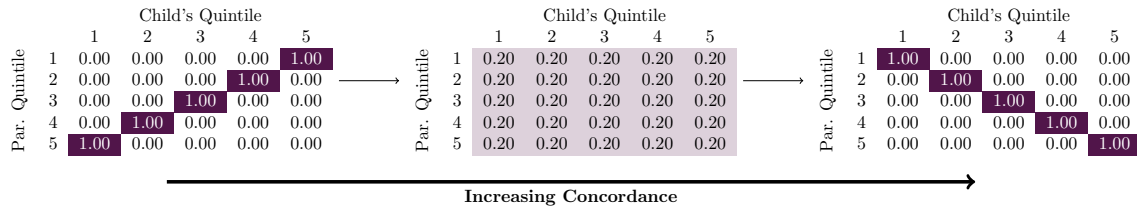


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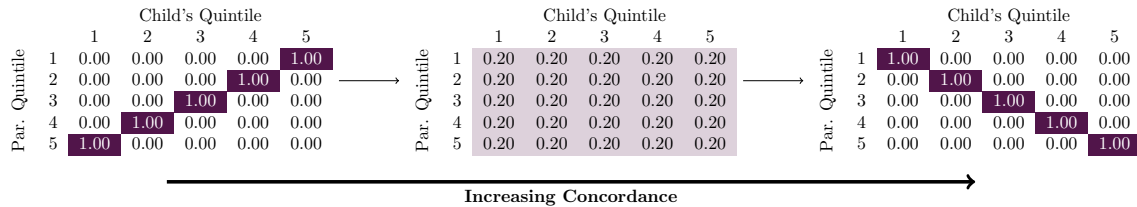
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- Pinned down by dependence parameters in parametric families (e.g., ρ in normal)

The Concordance Order

Definition: (Tchen 1980) Economy A is **more concordant** than economy B ($C^A \succeq C^B$) if:

$$C^A(u, v) \geq C^B(u, v) \quad \forall u, v \in [0, 1]$$

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 - “intuitive requirement that children from higher-income families are less likely to have lower incomes” (Chetty et al., 2017); “empirically unlikely transition matrices” (Shorrocks, 1978); “all-or-nothing” (Berman, 2022)
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 - Testable: provide transparent testing procedure
- Higher concordance $\longleftrightarrow$ **lower mobility**

Main Results: Sufficiency

Concordance Orders All Standard Mobility Measures

Props 1-4: Fix marginals. If $C^A \succeq C^B$, then economy A is *less mobile* than B by:

1. Regression/Correlation-based measures:
 - IGE: $\beta^A \geq \beta^B$, Rank-rank slope: $\rho^A \geq \rho^B$, Intergenerational correlation
2. “Local” measures:
 - Conditional expected rank $\longrightarrow$ rotates around $r^* = 0.5$
3. Transition probabilities:
 - diagonal $\uparrow$, off-diagonal $\downarrow$, Directional mobility
4. Axiomatic measures:
 - Shorrocks trace, Fields & Ok, D’Agostino & Dardanoni, Cowell & Flachaire

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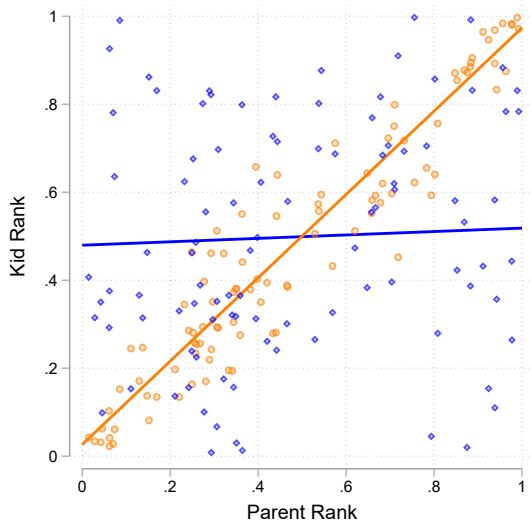
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$\rightarrow$ Measures are **ordinal substitutes**: observe one ranking, know all others

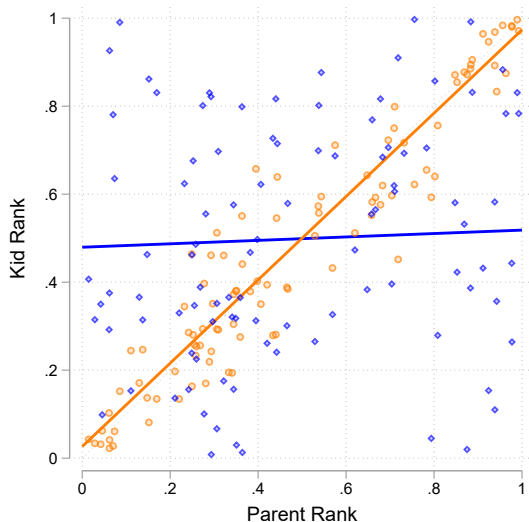
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Rank-rank slope



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Mobility Measure:

$$\rho = 12 \iint C(u, v) du dv - 3$$

$C^A \geq C^B$ everywhere

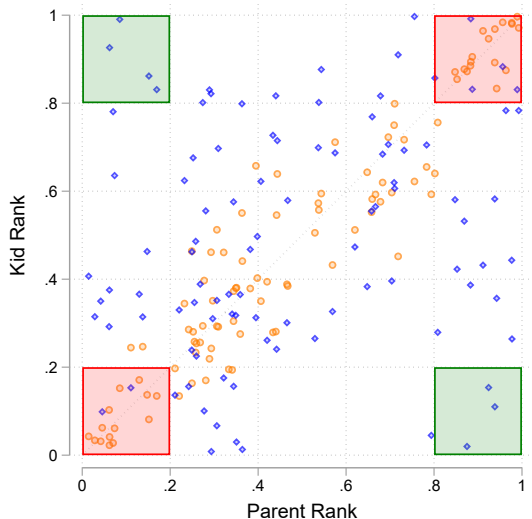
$$\longrightarrow \iint C^A \geq \iint C^B$$

$$\longrightarrow \rho^A \geq \rho^B$$

(Lehmann 1966; Tchen 1980)

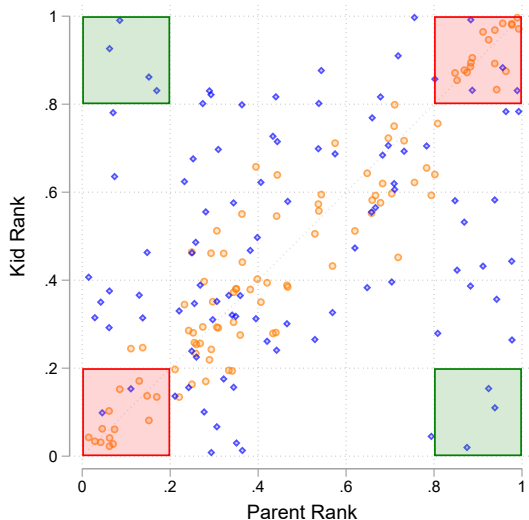
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Mobility Measure:

$$\Pr(R^K > \tau \mid R^P > \tau) = \frac{1 - 2\tau + C(\tau, \tau)}{1 - \tau}$$

If $C^A \geq C^B$ everywhere

$$\longrightarrow C^A(\tau, \tau) \geq C^B(\tau, \tau)$$

$$\longrightarrow \text{TP}^A \geq \text{TP}^B$$

(Copula definition, simplest cases!)



Concordance is Pigou-Dalton for Mobility

- Inequality measurement (Atkinson, 1970)
 1. Elementary operation: Pigou-Dalton transfer
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 3. Sufficiency: SOSD $\longrightarrow$ all transfer-sensitive measures agree

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- Mobility measurement (Today)
 1. Elementary operation: Concordance Exchanges
 - swap Y_i^K and Y_j^K if $Y_i^K < Y_j^K$, but $Y_i^P > Y_j^P \rightarrow$ increase concordance
 2. Partial order: Concordance
 - Discuss implications & completions
 3. Sufficiency: Concordance $\rightarrow$ all exchange mobility measures agree

Analogues in SWF for 2D ineq. (Atkinson-Bourguignon) & Perm. Income ineq. (Dardanoni)

U.S. Mobility Trends & Formal Test

Empirical Application: 20th Century U.S. Mobility

Datasets: Replicate **Jácome, Kuziemko & Naidu, 2025**

- Pool 15 U.S. surveys (ANES, GSS, PSID, OCG, NLS, ...) with respondent income + father's occupation (large sample of intergenerational links) 
- Impute parental income from Census using occupation $\times$ race $\times$ region (using census)
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Theory predicts:

- Rank-based measures should **co-move** , but other measures may diverge (rising inequality)

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Construct cardinal measure M_λ for (approx) concordance order

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$$M_\lambda = \lambda^* \in \operatorname{argmin}_{\lambda \in [0,1]} \int_0^1 \int_0^1 \left[C^{obs}(u, v) - \left((1 - \lambda) M(u, v) + \lambda C_{uv}^{\mathbb{I}} \right) \right]^2 du dv$$

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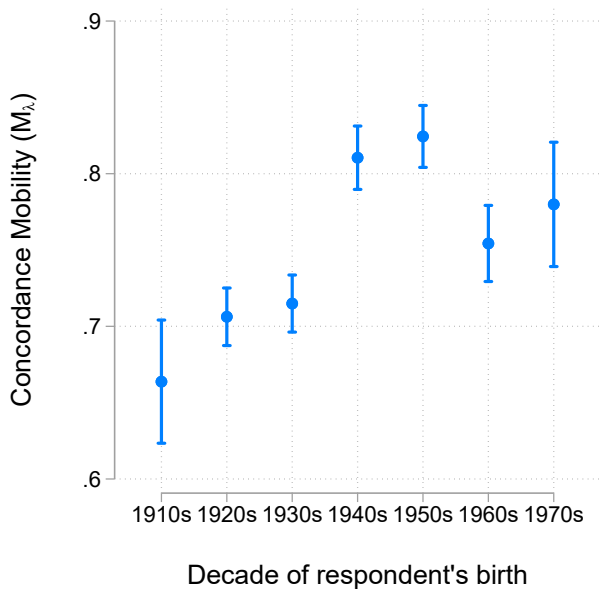
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Return to formal testing procedure

Concordance Mobility over the 20th Century



Wave-pattern

- Long run Mobility rises over the 20th century
- But plateaus/declines after 1960s

Increasing Mobility? Moment Inequality Test of Concordance Order

► Details

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1920	1930	.001	.004	Copulas cross
1930	1940	.052	.001	$C_{1930} \succeq C_{1940}$
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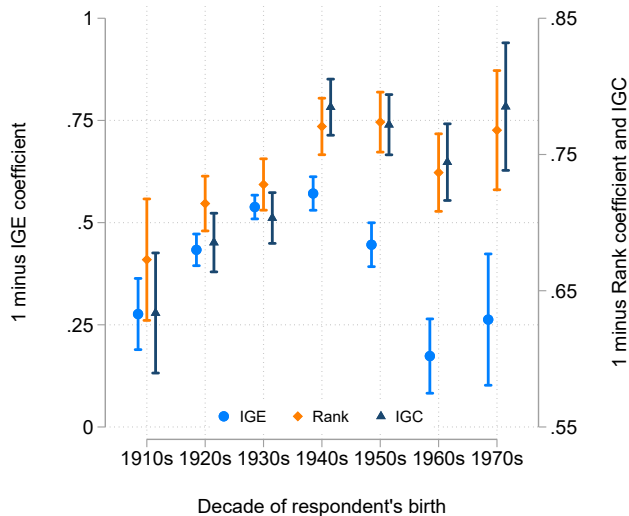
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- Some measure disagreement (paper has many measures) noise, others substantive
→ Transition Probabilities disagree when copulas cross

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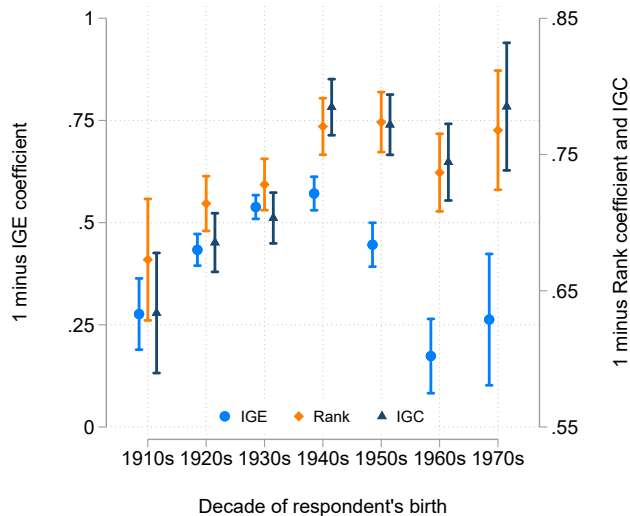
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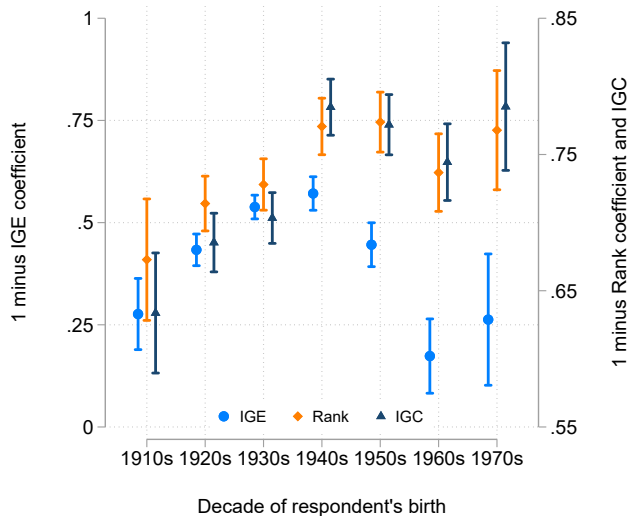
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- IGE diverges starting 1950s

Why? IGE adds changing inequality to theory results

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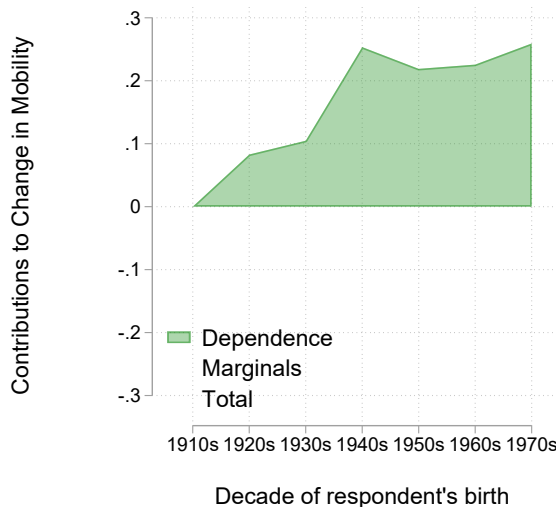
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- Level-based measures (IGE): Both components non-zero & potentially opposites
- Order matters $\longrightarrow$ average over fixing copulas at A and B
(Shapley-Owen-Shorrocks decomposition)

Decomposition: Why Does IGE Diverge?

► Other Measures



- **Dependence component:** rising mobility + plateau
 - Consistent with M_λ and R-R slope
 - Those are only dependence

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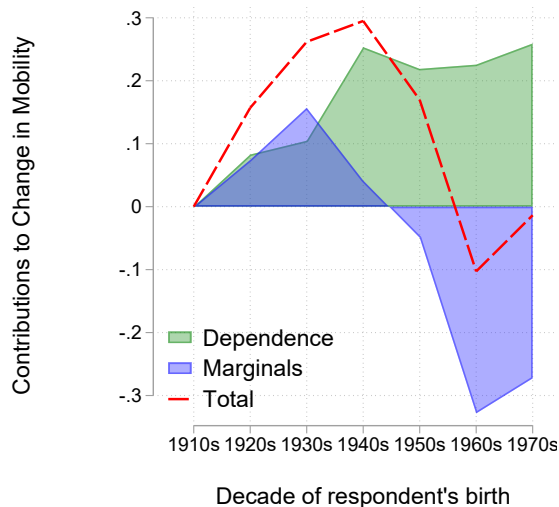


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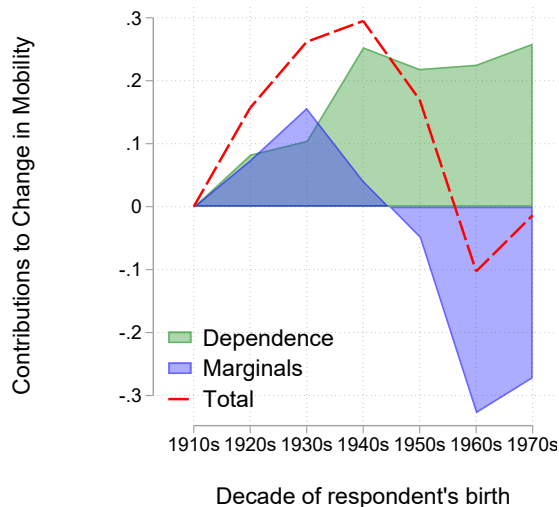
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- Disagreement reflects sensitivity to marginals!

What To Do When Concordance Fails?

Crossing Copulas: 1920s–30s and 1950s–60s ◀

Concordance is a partial order $\longrightarrow$ not all economies are ranked, e.g.

Par. Quintile	Child's Quintile				
	1	2	3	4	5
1	0.32	0.24	0.2	0.15	0.08
2	0.25	0.23	0.21	0.18	0.12
3	0.19	0.22	0.22	0.21	0.16
4	0.15	0.18	0.21	0.24	0.22
5	0.09	0.12	0.16	0.22	0.41

No
Concordance
Order
 $\longleftrightarrow$

Par. Quintile	Child's Quintile				
	1	2	3	4	5
1	0.58	0.00	0.00	0.14	0.28
2	0.00	0.86	0.00	0.00	0.14
3	0.00	0.00	1.00	0.00	0.00
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- Identical rank-correlation (0.32)

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- Identical rank-correlation (0.32)
- If economies not ranked by concordance, can always find two measures:
(Prop 3. & Corollary 2)
 - A is more mobile than B (1st meas.)
 - B is more mobile than A (2nd meas.)

Source of Disagreement

- Measures expressed as

$$M = \int \int \phi_M(Y^K, Y^P) f(Y^K, Y^P) dY^K dY^P,$$

where $\phi_M(Y^K, Y^P)$: mobility-measure specific intensity

- TP: Indicator; IGE: $1/\sigma_P^2(\ln Y^K - \ln \bar{Y}^K)(\ln Y^P - \ln \bar{Y}^P)$

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- Concordance guarantees $C^A - C^B \geq 0 \longrightarrow \Delta M \geq 0 \perp$ of weights 
- Failures $\longrightarrow C^A - C^B$ changes sign on diff. regions, but measures diff. sensitivity
 - Tie-break: **Normative** (intersecting Lorenz curves) or **robustness** choice (sensitivity to ϵ -violations)

What To Report When Concordance Fails

Prop 6: Concordance isn't knife-edge!

If $C^A - C^B$ within ϵ of satisfying concordance a measure M agrees up to $L_M \cdot \epsilon$

- L_M small for rank-based measures, larger for IGE and specific TPs
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Reporting protocol: use test output to localise the violation

- Report **local rank-rank slopes** on the regions where $C^A - C^B$ changes sign
- Supplement with transition probabilities at standard thresholds

Lets readers with different preferred measures reach their own conclusions

Other Settings & Structural Interpretation

No time? Jump to Conclusions

1. More measures (follow JEL taxonomy, Deutscher & Mazumder) include absolute measures
 - Lorenz mobility (McGee & Ocampo, 2025), Bartholomew jump measure, Goodman, average non-linear income persistence (Arellano, Blundell, Bonhomme, 2017), Katz & Krueger
2. Concordance order maintained under fixed rank comparisons
 - Ranking children by position in parent distributions
 - Subgroup or subnational regions against population distribution
3. Allow for multi-dimensional mobility
 - Whole lifecycle paths (Audoly, McGee, Ocampo & Paz-Pardo, 2025) or many generations

Microfoundation: Becker-Tomes-Loury

- Standard dynastic investment model

$$\max_{C^P, I} U(C^P) + \delta \mathbb{E}_\theta [U(C^K)]$$

subject to

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- Parents choose investment (I) and consumption (C^P); child consumes (C^K)
- $f(\cdot)$: skill production technology; increasing and (weakly) supermodular
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- θ : luck shock realized after investment
- When do changes in technology or preferences lead to more mobility?

Investment Expansions $\longrightarrow$ Larger in Concordance ▶ Example

Lemma 1: If investment expands uniformly ▶

$$I^{*A}(Y^P) \geq I^{*B}(Y^P) \quad \forall Y^P \longrightarrow C^A \succeq C^B$$

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What about **heterogeneous** policy impacts?

- E.g. school finance equalisation: poor districts $\uparrow$, rich districts $\downarrow$ (Biasi, 2023)
- $\Delta I^*(Y^P)$ crosses zero $\longrightarrow \Delta C$ changes sign $\longrightarrow$ **crossing copulas**
- Rationalises 1920s–30s, 1950s–60s?

Conclusions

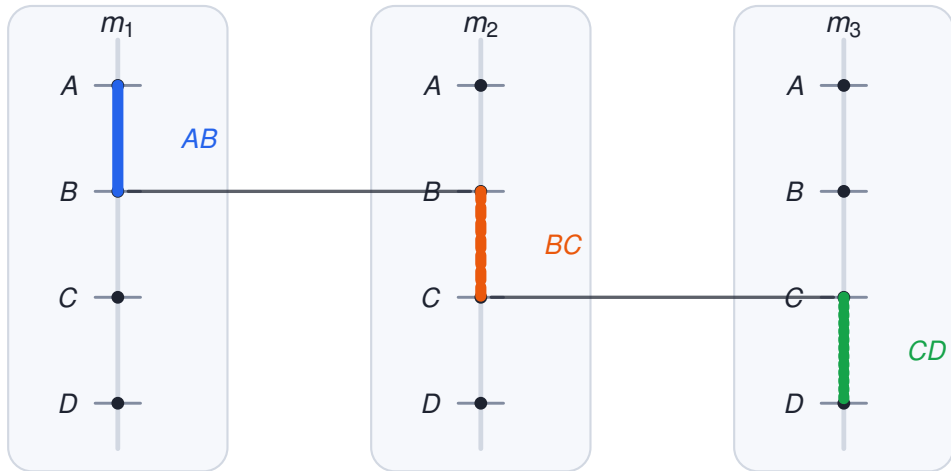
Implications For Applied Researchers

A single property—**concordance**—determines when standard mobility measures agree

For applied researchers:

1. **When concordance holds:** measures are substitutes
 - Pick by data quality, not axiom: transitions for noisy data, IGE for clean panels, rank-rank for cross-country
2. **When it fails:** disagreement is informative, not noise
 - Test it; report local rank-rank slopes; Lipschitz bounds for ϵ -violations
3. **Decomposition as diagnostic:** rank-dependence vs. changing inequality
(reframes IGE-vs-rank-rank divergence)

Chaining Measures



Gives full comparison of A, B, C and D in *any* measure

Details

e.g. from bunching, top-coding, censoring, genuine discreteness $\longrightarrow$ no restrictions on F_K or F_P , but Copula 'indeterminate'

Solutions:

1. Parametric copulas

- Does violence to discrete data
- Restrictive

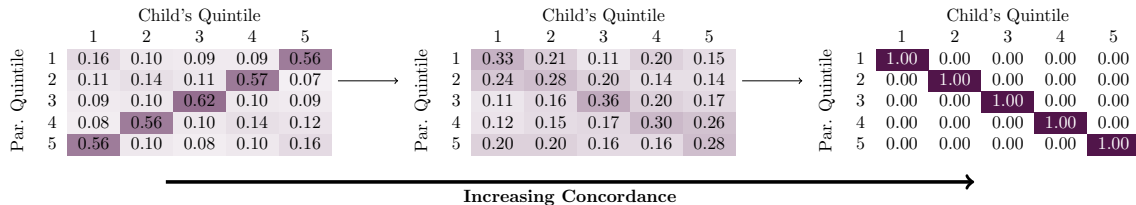
2. Carley (2002) bounds on copulas

- A lot more math, but flexible
- Worst-case bounds may not be tight

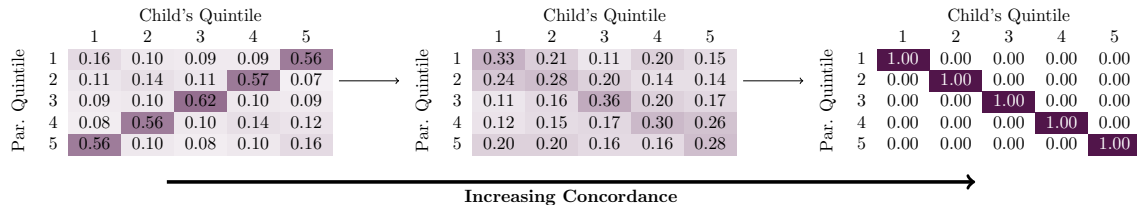
3. Pick an extension: checkerboard or Piecewise bilinear

- Propositions 11, 13 in Genest and Nešlehová (2007) show the unique extension preserving concordance order of the original joint-distributions

Concordance: Visual Intuition

[◀ Back](#)

Concordance: Visual Intuition [◀ Back](#)



- Not just extreme cases

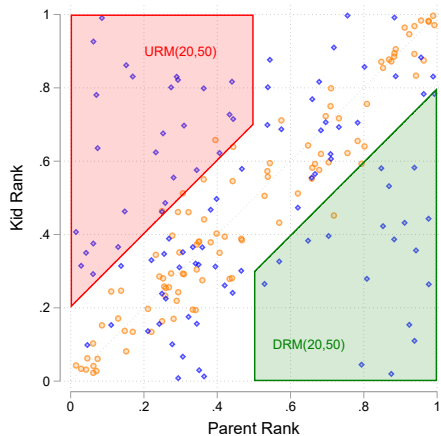
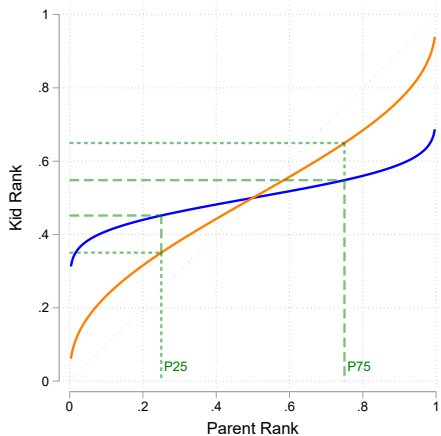
Some results require extra structure

1. Single crossing of conditional densities and continuity (sufficient: TP2 or MLRP)
 - For Non-parametric CER rotation
2. Smooth Anti-Diagonal to Diagonal Exchanges:

$$\Delta C(u_1, v_1) + \Delta C(u_2, v_2) \geq \Delta C(u_2, v_1) + \Delta C(u_1, v_2).$$

- Implies Concordance Order
- Directional Mobility & Shorrocks

Why Does Concordance Order These Measures? Illustration III

[Back](#)

Extension: Proxies & Measurement Error [◀ Back](#)

Prop 5: Observe proxy $E^g = H_g(R^g, \varepsilon^g)$ for generation g , H_g increasing, and ε independent of ranks. Let $\tilde{C}$ denote copula for (E^K, E^P) . **Concordance ordering preserved:**

$$C^A \succeq C^B \quad \longrightarrow \quad \tilde{C}^A \succeq \tilde{C}^B$$

Extension: Proxies & Measurement Error [◀ Back](#)

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2. Income and Education as skill proxies: $S \longrightarrow E, Y$

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1. Education/occupation as income proxies: $Y \longrightarrow E$

2. Income and Education as skill proxies: $S \longrightarrow E, Y$

- Covers flexible measurement error cases $\tilde{Y} = E = H_g(R^g, \varepsilon^g)$ (Corollary 1) 

1. (non-)Classical measurement error in (log-)incomes (Bound, Brown, Duncan & Rogers 1994)

2. Measurement error in ranks (Chetverikov & Wilhelm 2023)

3. Life cycle biases (Kitagawa, Nybom, & Stuhler 2018)

Proxies & Meas Error: Functional Forms [← Back](#)

Proxies:

1. Human capital determines incomes $Y = H(E) + \varepsilon$ with $E = H^{-1}(Y - \varepsilon)$
2. Ability determines education and income $E = H_E(S, \varepsilon)$ and $Y = H_Y(S, \varepsilon)$.

Proxies & Meas Error: Functional Forms [◀ Back](#)

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Measurement error:

1. Classical measurement error in incomes

$$\tilde{Y}^K = F_K^{-1}(R^K) + \varepsilon^K, \quad \tilde{Y}^P = F_P^{-1}(R^P) + \varepsilon^P$$

2. Measurement error in ranks

$$\tilde{R}^K = F_{\tilde{K}} \left(F_K^{-1}(R^K) + \varepsilon^K \right), \quad \tilde{R}^P = F_{\tilde{P}} \left(F_P^{-1}(R^P) + \varepsilon^P \right)$$

3. Life Cycle Biases

$$\tilde{Y}^K = G_K \left(F_K^{-1}(R^K) + \mu_{\varepsilon^K} + \sigma_{\varepsilon^K} \varepsilon^K \right), \quad \tilde{Y}^P = G_P \left(F_P^{-1}(R^P) + \mu_{\varepsilon^P} + \sigma_{\varepsilon^P} \varepsilon^P \right)$$

Measure specific weights [◀ Back](#)

- Measures expressed as

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where $\phi_M(Y^K, Y^P)$: mobility-measure specific intensity

- For fixed marginals, any twice-differentiable measure: $M_M^A - M_M^B$

$$= \int_0^1 \int_0^1 W_M(R^K, R^P; F^K, F^P) \left[C^A(R^K, R^P) - C^B(R^K, R^P) \right] dR^K dR^P$$

where

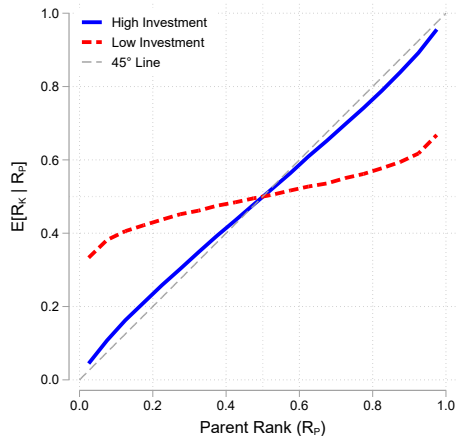
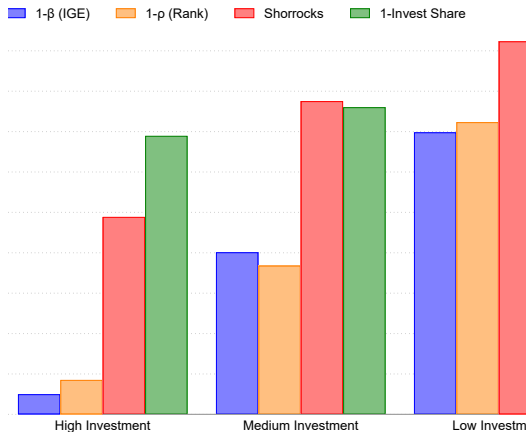
$$W_M(R^K, R^P) = \frac{\phi_{M,12}(F_K^{-1}(R^K), F_P^{-1}(R^P))}{f_K(F_K^{-1}(R^K)) f_P(F_P^{-1}(R^P))}$$

- Works for non-continuous cases also

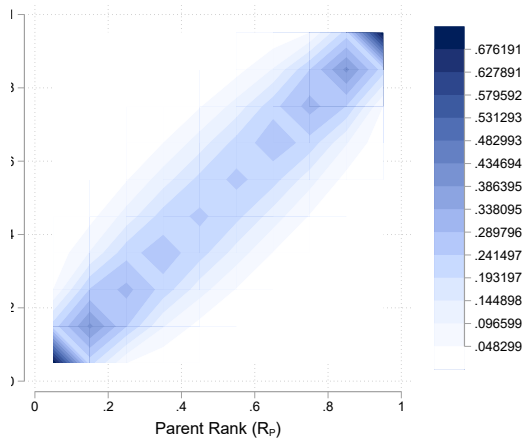
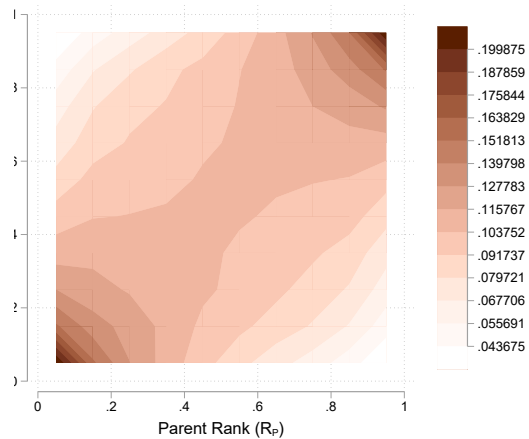
Investment Expansions Example

[◀ Back](#)

Log-utility and Cobb-Douglas Production



Investment Expansions Example

[◀ Back](#)

Proof Sketch: Lemma (Investment $\longrightarrow$ Concordance) [◀ Back](#)

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$$H^{K,A} \mid Y^P = y \succeq_{\text{FOSD}} H^{K,B} \mid Y^P = y \quad \forall y$$

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- But CDF not common, 3 cases:
 1. *Rank-dependent wages*: Integrate over $Y^P \Rightarrow (R^K, Y^P)^A \succeq (R^K, Y^P)^B$
 2. *Common wage function*: FOSD of child incomes + decomposition theorem $\longrightarrow$ concordance $\uparrow$
 3. *Increased returns*: Analogous reasoning applies

Proof Sketch: Corollary (Policy $\longrightarrow$ Investment) [◀ Back](#)

Define value function for investment I and 'policy' λ :

$$V(I; \lambda) = U(C^P) + \delta_\lambda \mathbb{E}_\theta[U(W_\lambda(H^K))]$$

subsume subsidy or transfer into f

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Key condition (Topkis): Increasing differences holds if

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Verification:

- Production function f is supermodular
- Wage schedule W_λ is strictly increasing
- Standard regularity on GE effects (e.g., multiplicative shifts)

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- Production function f is supermodular
- Wage schedule W_λ is strictly increasing
- Standard regularity on GE effects (e.g., multiplicative shifts)

Conclusion: $\lambda' < \lambda \implies I^*(Y^P; \lambda') \leq I^*(Y^P; \lambda)$

Jácome, Kuziemko & Naidu (2025) Replication [◀ Back](#)

15 Pooled Surveys:

- General Social Survey (GSS), American National Election Survey (ANES)
- Panel Study of Income Dynamics (PSID), Occupational Changes in a Generation (OCG)
- National Longitudinal Surveys (NLS): Mature Women, Older Men, Younger Women
- National Survey of Black Americans, National Fertility Survey, others

Sample Construction:

- Inclusion: US-born, ages 30–50, non-missing family income + father's occupation
- Race: White and Black only (other races < 1% pre-1950)
- Weights: Reweighted to census race $\times$ sex shares per birth decade

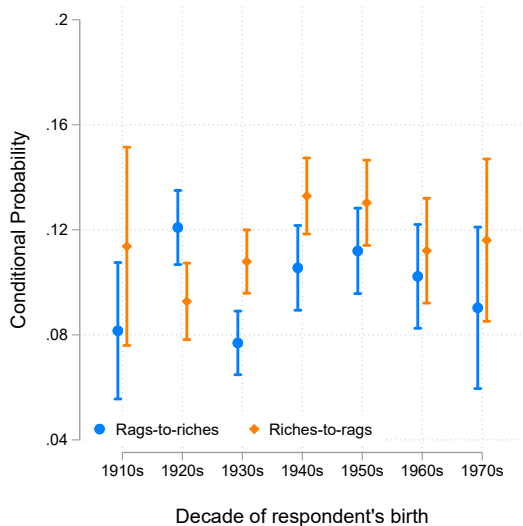
Parental Income Imputation:

- Predict $\hat{Y}^P$ using 28 harmonized occupation categories $\times$ race $\times$ South
- Sources: 1901 Cost of Living Survey, 1936 Expenditure Survey, 1940–90 Censuses
- Matched to respondent's $\approx$ 10th birthday; special adjustments for farmers

Sample Sizes by Birth Decade:

1910s	1920s	1930s	1940s	1950s	1960s	1970s
4,641	11,857	11,426	10,288	9,531	5,691	2,668

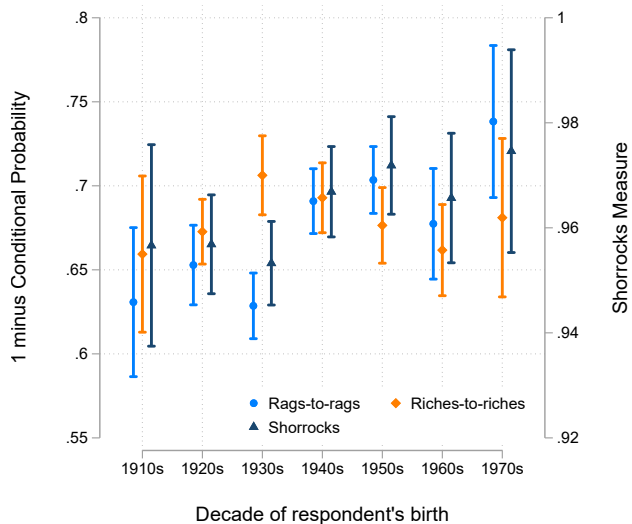
Transition Matrix Measures Co-Move

[◀ Back](#)

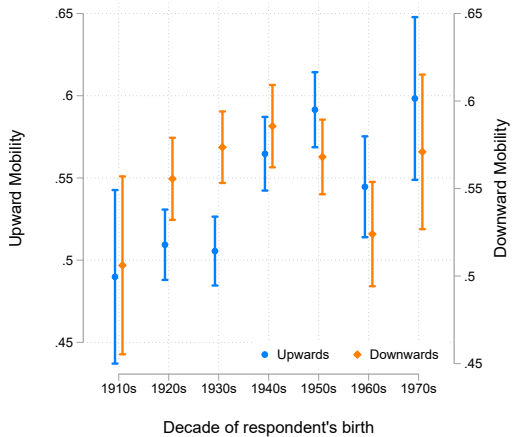
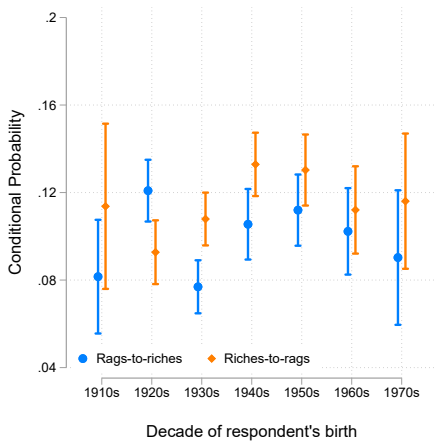
- Quintile Transition Matrix

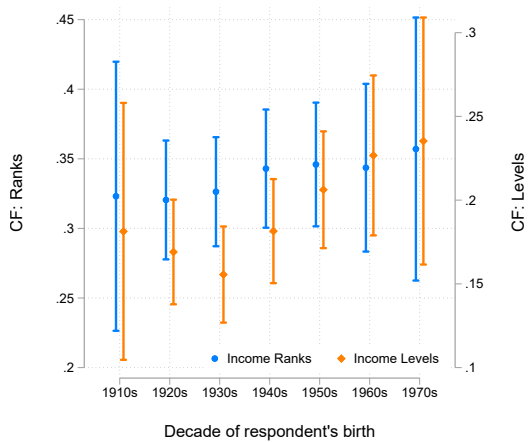
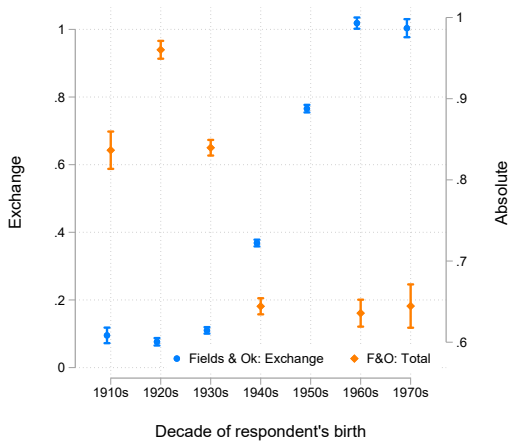
- Q1→Q5 & Q5→Q1

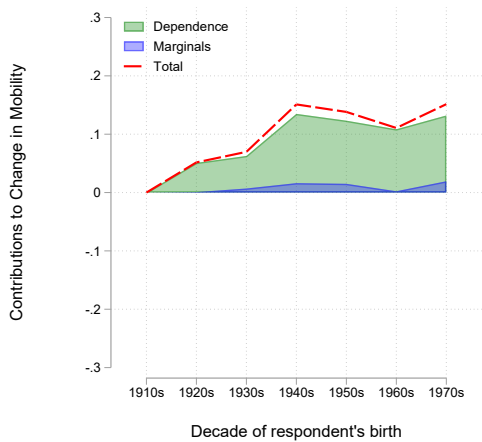
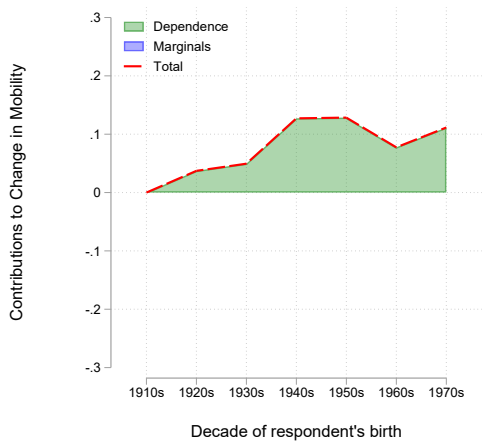
Transition Matrix Measures Co-Move [◀ Back](#)

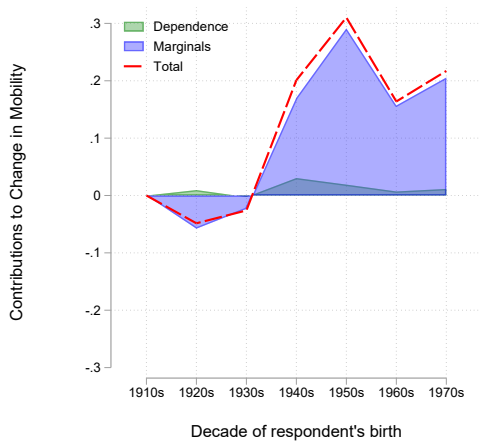
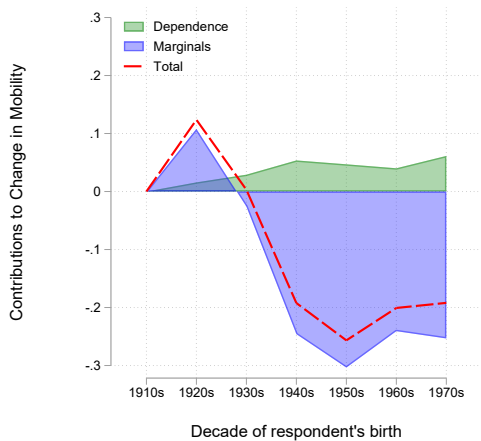


- Quintile Transition Matrix
- Q5→Q5 & Q1→Q1 & Trace









Testing The Concordance Order [◀ Back](#)

Set up as a moment inequality testing procedure (Andrews and Soares, 2010)

- Moments on a grid $\mathcal{G}$:

$$m_g = C_A(u_g, v_g) - C_B(u_g, v_g), \quad g = 1, \dots, G.$$

- Null hypothesis of dominance

$$H_0: m_g \geq 0 \text{ for all } g = 1, \dots, G \quad \text{vs.} \quad H_1: m_g < 0 \text{ for some } g.$$

- Test statistic is the minimum of the studentized moments:

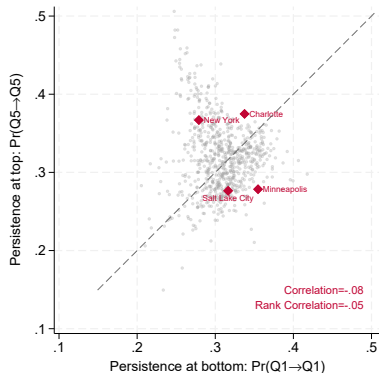
$$T_n = \min_{g=1, \dots, G} \frac{\hat{m}_g}{\hat{s}_g},$$

w/ bootstrap standard error $\hat{s}_g$

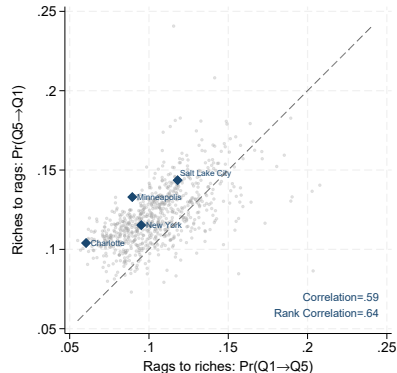
- Test $A \succeq B$ and $B \succeq A$

Spatial Mobility: Measure Disagreement Across CZs [◀ Back](#)

Chetty et al. (2014) CZ transition matrices, remapped to within-CZ copulas



$Q1 \rightarrow Q1$ vs. $Q5 \rightarrow Q5$ persistence

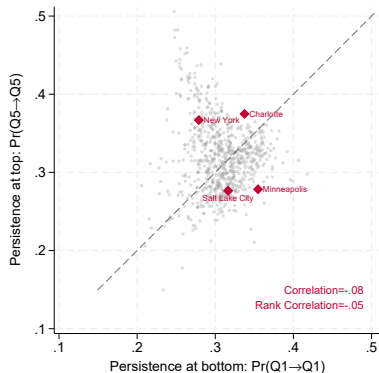


$Q1 \rightarrow Q5$ vs. $Q5 \rightarrow Q1$ tail mobility

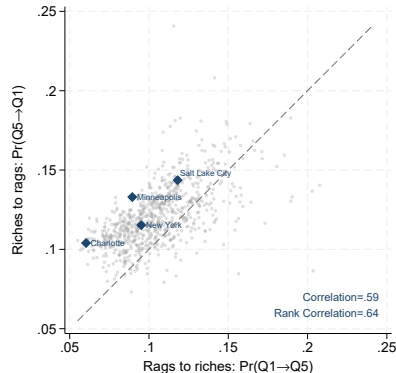
- Salt Lake $\succ$ Charlotte on *every* measure...
 - ...but Salt Lake vs. NYC, NYC vs. Minneapolis: depends on measure

Spatial Mobility: Measure Disagreement Across CZs [◀ Back](#)

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$Q1 \rightarrow Q5$ vs. $Q5 \rightarrow Q1$ tail mobility

- Salt Lake $\succ$ Charlotte on *every* measure...
 - ...but Salt Lake vs. NYC, NYC vs. Minneapolis: depends on measure
- Persistence top-quartile $\neq$ in **91%** of CZs; **61%** for the tail-mobility

A Simple Illustrative Example [◀ Back](#)

Separating Dependence & Inequality

Assume Incomes jointly Log-Normal (closed form)

$$\begin{pmatrix} \ln Y^K \\ \ln Y^P \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_K \\ \mu_P \end{pmatrix}, \begin{pmatrix} \sigma_K^2 & \rho\sigma_K\sigma_P \\ \rho\sigma_K\sigma_P & \sigma_P^2 \end{pmatrix} \right]$$

A Simple Illustrative Example [◀ Back](#)

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A Simple Illustrative Example [◀ Back](#)

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A Simple Illustrative Example [◀ Back](#)

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- IGE: $\beta = \rho \cdot \frac{\sigma_K}{\sigma_P}$ *dependence and inequality*

$$\begin{aligned} \text{- Changes: } \Delta\beta = & \underbrace{(\rho^B - \rho^A) \cdot \frac{\sigma_K^B}{\sigma_P^B}}_{\text{change in rank correlation}} + \underbrace{\rho^A \cdot \left(\frac{\sigma_K^B}{\sigma_P^B} - \frac{\sigma_K^A}{\sigma_P^A} \right)}_{\text{change in inequality}} \end{aligned}$$

A Simple Illustrative Example [◀ Back](#)

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- Rising inequality ($\sigma_K/\sigma_P \uparrow$) can **dominate** IGE even as rank dependence **weakens** ($\rho \downarrow$)